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From Leaderboard Ranks to Audited Decisions: A Controlled-Panel Study of Evaluation-Format Uncertainty

Anonymous ACL submission

Abstract

A persistent evaluation gap in model leaderboards is that rank tables often hide the pairwise, scorer, metric, and sampling choices that determine whether one model can be responsibly described as better than another. Price work makes benchmark and prompt sensitivity visible, but it leaves a practical author-facing question open: after a fixed set of official scorer outputs has been collected, which pairwise model claims should a paper state, soften, or withhold? We introduce a controlled-panel protocol that converts those outputs into stable-or-abstain reports within each benchmark family and metric, a model pair receives a direction only when the chosen confidence-bound rule supports it, and otherwise the paper records abstention. We evaluate the audit on 2560 scored rows for 10 open instruction/chat models across 5 benchmark families, with same-output baselines that approximate prompt reruns/decision, rank-interval uncertainty, and benchmark-only shortcuts. Under the default rule, 15 of the 300 simultaneous bands yield 15 stable decisions and 300 abstentions over 315 model-pair/family/metric comparisons, while naive intervals would report 45 stable decisions and the observed evidence-format mix produces 53 direction-flipping pairs. These findings turn uncertainty into a reporting standard for the controlled panel: publish directional claims when the interval rule authorizes them, and otherwise state abstention and the missing evidence needed for stronger public-leaderboard claims.

1 Introduction

Public leaderboards compress benchmark choice, prompt or formatting convention, scorer behavior, and metric aggregation into a single ordering. That compression is useful for scanning models, but it creates an evaluation gap when a practitioner treats an adjacent rank as a reliable pairwise recommendation. In that setting, the scientific question is not only where a model sits in an ordered list; it is whether a paper should turn a small score gap into a directional sentence about one model being better than another.

Prior work gives strong reasons to be cautious. Benchmark-targeting studies show that rankings can move under prompt and scoring perturbations (Alzahrani et al., 2024a,b), and efficient multi-prompt evaluation and rank-uncertainty methods give tools for aggregating or internalizing such variation (Polo et al., 2024b; Neuhaus and Benjamini, 2026; Avdac Menendez et al., 2026). Those methods make uncertainty visible, but a manuscript still needs an author-facing decision rule after its scorer tensor is fixed: state a direction, soften it, or withhold it.

We therefore treat leaderboard validity as a claim-reporting problem rather than as another rank-correlation metric. A rank table always orders systems, even when score differences are inside the noise induced by task sampling, evaluator settings, or metric choice; a reporting decision can instead say "we do not know." Earlier NLP evaluation work has called for stronger reporting of variance, significance, and experimental conditions (Dror et al., 2018; Dodge et al., 2019; Card et al., 2020). Our audit makes that discipline operational: it starts from official scorer outputs, forms model pairs within each benchmark family and metric, applies pairwise and simultaneous confidence intervals, and labels each pair as stable or abstained. A separate evaluator-format audit checks whether the observed chat-template versus no-chat axis changes the pair direction, so sensitivity matters when the claim without being mistaken for a certified public-leaderboard reversal.

The empirical question is whether this stable-or-abstain rule changes what the paper should report. The completed run covers MMLU (Hendrycks et al., 2021), ARC-Challenge (Clark et al., 2018), HellaSwag (Zellers et al., 2019), GSM8K (Cobbe

Evaluation reliability

Audited leaderboard decisions

15 stable · 300 abstained

When Structure Does Not Help: A Diagnostic Study of Training-Free Gates for Compositional Image–Text Matching

Anonymous ACL submission

Abstract

Compositional image–text matching benchmarks ask whether a vision–language model binds attributes, relations, and word order rather than merely recognizing the right objects. Because caption-to-atom QA is already close to TIFA, Davidsonian scene graphs, and recent training-free matching work, we test a narrower reliability hypothesis: dependency-aware atom evidence should act only as a selective veto around holistic VQAScore, with OpenCLIP admitted when the holistic margin is weak. The fixed pairwise protocol scores each image against the benchmark positive and hard-negative captions, and the gate $\tau = 0.942$ (train-set tensor is fixed: state a direction, soften it, or withhold it.

We therefore treat leaderboard validity as a claim-reporting problem rather than as another rank-correlation metric. A rank table always orders systems, even when score differences are inside the noise induced by task sampling, evaluator settings, or metric choice; a reporting decision can instead say "we do not know." Earlier NLP evaluation work has called for stronger reporting of variance, significance, and experimental conditions (Dror et al., 2018; Dodge et al., 2019; Card et al., 2020). Our audit makes that discipline operational: it starts from official scorer outputs, forms model pairs within each benchmark family and metric, applies pairwise and simultaneous confidence intervals, and labels each pair as stable or abstained. A separate evaluator-format audit checks whether the observed chat-template versus no-chat axis changes the pair direction, so sensitivity matters when the claim without being mistaken for a certified public-leaderboard reversal.

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Vision--language matching

Compositional matching gate

0.953 fallback > 0.942 gate

When a Frozen Gate Refuses to Adapt: A Diagnostic Audit of Test-Time Adaptation in Vision-Language Models

Anonymous submission

Abstract

Training-free test-time adaptation can fail though the controller that decides which updates enter persistent model state. We study this failure mode in a frozen-gate DOTA setting for a frozen-risk gate around DOTA updates for OpenCLIP ViT-B/16 on DTD, ImageNet-A, and Oxford-IIIT Pet. At each stream/order/seed, DOTA proposes new class statistics, while an unlabelled live component score measures the evidence for each update. The gate admits updates with probability or soft-back the state change or threshold 0.0000. The gate accepts only 2/1,880 DTD updates and 8/13,669 Oxford-IIIT Pet updates, and its ImageNet-A behavior is statistically bounds below 10%. In all three families, gated top-1 accuracy is closer to unadapted OpenCLIP than to native DOTA, while ECE and negative log-likelihood are significantly worse. Thus the fixed gate suppresses native adaptation rather than providing a reliable middle ground. Our diagnostic evaluates lightweight adaptation gates as state controllers, with acceptance intervals, constant-all controls, and paired outcome deltas across multiple public benchmark families.

Introduction

Vision-language models such as CLIP provide strong zero-shot classifiers without task-specific training (Radford et al., 2021; Zhai et al., 2023). Deployment shifts nevertheless can degrade their performance, and test-time adaptation has begun to mature: style, corruption, and class retrieval (Dong et al., 2009; Hendrycks et al., 2021; Hendrycks and Dietterich, 2019; Recht et al., 2019). Existing methods update prompts, caches, normalization statistics, or model state using entropy, agreement, confidence, or pseudo-label criteria (Sun et al., 2020; Wang et al., 2021; Zhang, Levine, and Finn, 2021; Liang, He, and Tan, 2025). Our audit inherits this concern but focuses on a single controller interface: whether a frozen gate should commit a proposed DOTA update.

This omission matters for stateful adaptation. Recent VLM methods use cache admission, augmentation consistency, or multi-prompt agreement to decide when updates help (Karmanov et al., 2024; Zanella and Ben Ayed, 2024; Farina et al., 2024; Zhou et al., 2025; Zanella et al., 2025). DOTA maintains distributional state across the stream (Han et al., 2025). A gate can protect such state by rejecting risky proposals, but excessive rejection can instead erase the benefit of adaptation. Separating these cases requires the re-update trajectory, the counter-all adaptive trajectory, and the gated trajectory on the same examples.

We compare these three trajectories for OpenCLIP ViT-B/16 (Ilharco et al., 2021) on DTD, ImageNet-A, and Oxford-IIIT Pet streams. The frozen threshold is 0.0004. It accepts 2/1,880 DTD updates, 1,932/7,500 ImageNet-A updates, and 8/13,669 Oxford-IIIT Pet updates. In all three families, gated top-1 accuracy is closer to unadapted OpenCLIP than to native DOTA, with gated–DOTA differences of −0.0188/[−0.0188/[−0.0188, −0.0120], and −0.0120/[−0.0120, −0.0120], respectively.

The study makes three contributions. First, it evaluates a lightweight gate as a state controller through acceptance intervals and paired outcomes, rather than through accuracy alone. Second, it shows that the frozen gate's behavior in DTD and Oxford-IIIT Pet while showing a distinct refusal-dependent on ImageNet-A. Third, it tests the proposed risk interpretation directly and finds weak or inverted harmful-update ranking. These conclusions apply to the fixed controller, backbone, and benchmark streams evaluated here, not to update-gating or DOTA in general.

Related Work

Test-time adaptation and online state. Test-time training and fully test-time adaptation update a deployed model using only test inputs, commonly by self-supervised or entropy-minimization objectives (Sun et al., 2020; Wang et al., 2021). Continual and dynamic variants add memory, resets, or stochastic restoration to limit drift (Wang et al., 2022; Niu et al., 2022; Hendrycks and Dietterich, 2019; Recht et al., 2019). Surveys emphasize that source-free adaptation can be sensitive to stream order, pseudo-label quality, and hidden assumptions about test distributions (Liang, He, and Tan, 2025). Our audit inherits this concern but focuses on a single controller interface: whether a frozen gate should commit a proposed DOTA update.

Vision-language test-time adaptation. Prompt-learning methods such as CoOp and CoCoOp adapt text prompts to changing test representations or prompts (Zhou et al., 2022b,a; Shu et al., 2022). Recent training-free or lightweight alternatives use caches, augmentation consensus, weight averaging, calibration constraints, Bayesian priors, retrieval, exploration, or dependence-adjustment mechanisms (Karmanov et al., 2024; Zanella and Ben Ayed 2024;

Test-time adaptation

Frozen adaptation gate

2 / 1,880 DTD updates

Surface-Prior Selection for Training-Free GUI Reliability at Matched Coverage

Anonymous ACL submission

Abstract

GUI agents are usually evaluated as predictors, but deployment risk depends on which wrong actions a wrapper lets through. We study a training-free reliability wrapper around a frozen Qwen2.5-VL candidate pool and ask whether it can delay high-cost wrong executions without using calibration data, fitted thresholds, or parameter updates. The central device is matched coverage: for every benchmark family and coverage level, all selectors admit the same number of rows before ordering them by a surface prior, isolating selection quality. We propose a surface-prior selector that ranks by fixed action-cost labels and breaks tied rows with public rank-surface cues such as absent web-candidate-set size, no-provision actions, short instructions, and text/icon grounding actions. On 300-row active slices from ScreenSpec, GUIOdyssey, MMBench-GUI, and MiniCPM4b, this selector reduces aggregate high-cost wrong-execution area from 0.0125 for Ask-before-act to 0.0115, with paired baseline-minus-ours delta 0.0010 and 95% CI [0.0003, 0.0017]. The result is useful but bounded: the surface prior is not universal, on one of four active families, task recovery is not established, and MiniCPM4b action F1 must remain separate from exact-step success.

1 Introduction

GUI agents increasingly operate directly on screenshots, mobile screens, web pages, and desktop interfaces. This shift is visible in screenshot-based grounding models such as SeeClick and CogAgent (Cheng et al., 2024; Hong et al., 2024), high-resolution professional GUI benchmarks such as ScreenSpot-Pro (Li et al., 2025), coordinate-free and parser-based grounding systems such as GUI-Actor and OmniParser (Wu et al., 2025; Lu et al., 2024b), and broader UI understanding models such as ScreenAI and Ferret-UI (Baechler et al., 2024; You et al., 2024). These systems make GUI control look like an ordinary prediction problem, but the evaluation problem changes once predictions become actions. A wrong click on a harmless menu, a wrong click on a destructive confirmation button, and a partially correct web-action are different reliability events even when they are represented as one row in a benchmark table.

The natural response is to wrap a frozen agent with a selective execution rule: execute when the action looks safe and abstain or ask when it looks risky. This strategy connects GUI automation to older reject-option classification (Chow, 1970; Geifman and El-Yaniv, 2017), calibration and confidence estimation (Guo et al., 2017; Minderer et al., 2021), and more recent conformal or risk-controlling prediction sets (Angelopoulos and Bates, 2021; Bates et al., 2024; Angelopoulos et al., 2024). In GUI agents, however, a wrapper can look safer for reasons that do not establish safer execution. It may simply execute fewer examples, rank easy rows first, report action-overlap instead of exact step success, or compare against a published baseline outside that baseline's comparable task family. A reviewer then cannot tell whether the safety claim comes from better ordering of risky actions, lower coverage, a different scorer, or an inapplicable comparison.

This paper asks a narrower question: under identical coverage, can a training-free wrapper around a frozen GUI candidate pool reduce high-cost wrong executions relative to a strong non-learned baseline? The evaluated systems sits outside Qwen2.5-VL (Qwen Team, 2025); it trains no model, uses no calibration set, fits no threshold, and only reorders fixed candidate records. Matched coverage is the central methodological device. Within each benchmark family and coverage target, every method executes the same number of rows, so abstention is self cannot be the hidden intervention. The method uses action-cost labels as the main ordering signal and breaks otherwise tied rows with a tiny surface

Multimodal agents

GUI reliability at matched coverage

0.0125 → 0.0115 risk area

When Frozen Multimodal Hallucination Wrappers Fail

Anonymous submission

Abstract

Training-free wrappers promise a cheap way to reduce hallucination in frozen multimodal language models, yet their evaluations rarely ask whether they improve more than a baseline ever wins. We audit calibration-first wrappers for selected 3B-scale POPE, AMBER, and HallusionBench subsets with a frozen Qwen2.5-VL-3B-Instruct backend and official scorer outputs. The retained matrix does not include a no-wrapper baseline, so the audit is limited to rerunning and evidence-comparison arbitration on bounded official answers. NoLan fixed-alpha decoding has small retained-parser deltas on POPE (+0.7 points), and retained-parser deltas on HallusionBench (+1.7 points), but its AMBER delta is −0.3 points, with a 95% bootstrap CI [−0.8, +0.1] that crosses zero. Its mean decoding time per example is 1.5× the retained baseline. Our evidence-aware arbitration wrapper is conservative: it changes 0.0–0.3% of rows, and its AMBER CI [−0.2, +0.1] also crosses zero. Under a 0.5-point practical-improvement threshold, no wrapper in the audited slice is practically better than baseline. The contribution is a reproducible audit protocol: official scorer outputs, retained-parser reruns, bootstrap intervals, and explicit practical thresholds can prevent small wrapper gains from being overinterpreted.

1 Introduction

Multimodal hallucination work has moved from caption-only object errors toward object, attribute, relation, and visual-question failures in large vision-language models. Early captioning analyses established that models can name salient objects while inventing unsupported details (Rohrbach et al., 2018). Later POPE, AMBER, HallusionBench, CHAIR, TIFGEN, triplet-level evaluation, and RecHeart broaden the target to discriminative object existence, relation, and entangled visual-language reasoning (Li et al., 2023; Wang et al., 2023; Guan et al., 2024; Kaul et al., 2024; Wu et al., 2024; Zhang et al., 2026). The practical pressure is that a wrapper can look useful by changing only a few rows, so a large neutral-line studies decoding-time, attention, visual-contrast, grounding, and correction mechanisms that aim to suppress hallucinated answers without changing the base model weights (Leng et al., 2024; Huang et al., 2024; Chen et al., 2024; Yang et al., 2025; He et al., 2025; Li et al., 2025; Zhang and Yang 2025).

That attraction creates an evaluation trap. A wrapper can add a verifier prompt, a contrastive score, or an evidence comparison around a frozen backend while still leaving retained official answers unchanged, or it can change many answers in a direction that hurts one benchmark and only weakly helps another. Recent 2026 proposals sharpen this risk: they report gains from visual-token pruning, hallucination-aware scoring, and evidence-aware arbitration, but the strongest claims rely on small subsets, token-level visual sensitivity, or tool-free attributions (Anand et al. 2026; Li et al. 2026b; Qu et al. 2026; Tang et al. 2026; Lei et al. 2026; Guo et al. 2026; Zhang et al. 2026; Tsvi et al. 2026). A negative audit of such wrappers is therefore useful only if it separates three states: no retained harm, a small harm, non-decisive gained movement, and decisive degradation.

The separation is especially important because the newer intervention families differ in what they claim to perturb. Some target visual returns or localized attention, some start heads or reduce language-prior dominance, and others introduce attributions, capture guidance, or causal decoding (Chen et al. 2024; Huang et al. 2024; Yang et al. 2025; He et al. 2025; Yue et al. 2026; Chang et al. 2026b; Chang et al. 2026; Huang, Zhang, and Luo 2026; Yu et al. 2026). A mean-score table alone cannot tell whether such mechanisms created a legitimate visual correction, a harmless formatting change, or a benchmark-specific shortcut. A row-level audit can force more answer identity as a first-class outcome. If the official-answer string is unchanged after normalization, the wrapper is counted as a no-op for the retained scorer even when its intermediate controller output differs. If an answer changes, the audit separates whether the change fixes or harms paired correctness.

This paper asks a deliberately narrow question under a retained public-benchmark matrix, when do calibration-first wrappers around a frozen multimodal model produce real official-answer changes, and when do they merely look like they might help? Figure 1 summarizes the design: official scorer outputs are the primary evidence, a frozen prompt baseline on three benchmark families, using official scorer outputs and paired row tests rather than unpaired score summaries. The contribution is not a new positive mitigation method. It is a failure-mode audit for the kind of wrapper claim that can otherwise survive aggregate-score comparison: official answers are re-scored under identical to the frozen baseline across POPE, AMBER, and HallusionBench. Second, NoLan fixed-alpha decoding has small retained-parser deltas on POPE (+0.7 points), AMBER (+0.3 points), and HallusionBench (+1.7 points), all intervals span zero, and a strict invalid-to-wrong sen-

Multimodal hallucination

Frozen hallucination wrappers

15 cells · 4,500 rows

When Vocabulary-Row Proxies Fail: A Negative Result for Calibration-Free Vocabulary Quantization

Anonymous submission

Abstract

Compressing the embedding and output projection matrices of large vocabulary language models is attractive, but it is not an autoregressive deployment gap: practical quantizers already assign these tensors low-bit formats. We therefore test a narrower hypothesis: whether a completely calibration-free selector, using only tokenizer statistics, public token frequencies, row norms, and public entity-like membership, can choose high-precision vocabulary rows that preserve rare factual behavior better than random row buckets under the same vocabulary-byte budget. On a completed Qwen3-4B matrix with 1,000 rows per cell, TriviaQA, Natural Questions rare-entity QA, MMLU, ARC-Challenge, and HellaSwag, the answer is negative. The best selector diagnostics are numerically close to random bypass controls: TriviaQA exact match is 41.00% versus a 40.73% random mean, Natural Questions is 13.00% versus 13.20%, and the tie-aware macro average is 72.22% versus 71.90%. Paired sign tests yield 0 significant random-control wins at p<0.05, while WikiText-2 perplexity has only one significant win out of 20 comparisons. The result does not make vocabulary quantization unimportant; it shows that public-frequency/entity-like proxies are too weak to support a positive low-bit factuality claim without calibration or stronger evidence.

1 Introduction

Vocabulary matrices are a visible memory component in modern decoder-only language models. The input embedding and output projection can scale with vocabulary size, and models such as Qwen3-8B are deployed through frameworks that already expose memory/latency tradeoffs for both dense inputs and quantized variants (Qwen Team, 2025; Wolf et al., 2020; GGML community, 2023). This makes the naive premise that embedding and output heads are ignored by deployment quantizers untenable. The useful question is instead narrower: when those matrices are compressed under a fixed byte budget, can the rows that deserve extra precision be identified without forward calibration?

This paper studies that question as a failed-proxy experiment rather than as a success story. The intended mechanism is deliberately austere. It sees the tokenizer, public frequency counts, public entity-like token membership, and static row norms from the vocabulary matrices. It does not inspect validation answers, model predictions, loss traces, activation statistics, or calibration forward passes. That design makes the selector operationally cheap and leakage-resistant, but it also forces the core assumption into the open: high-frequency or high-norm entity-like rows must coincide with rows that matter for long-tail factual recall.

The distinction matters because vocabulary-row selection is unusually easy to overinterpret. A row can look rare, entity-like, or geometrically salient while still being irrelevant to the exact answer token. Tokenization used by a QA benchmark. Conversely, an answer can depend on a common subword, a punctuation-sensitive continuation, or a decoder state that is not localized to one protected vocabulary row. A calibration-free policy is attractive precisely because it avoids forward passes, yet that same restriction removes the strongest evidence that a row actually changes model behavior. The experiment therefore treats public metadata as a hypothesis about row importance, not as a substitute for behavioral validation.

The completed evidence rejects the broad version of that assumption. Across 18,834 scored rows from six raw benchmark families plus a derived tie-aware macro view, the strongest selector diagnostics do not consistently separate from random bypass controls. The most favorable rare-entity row reaches 41.00% exact match on TriviaQA versus a 40.73% random control mean, but the same entity-title plus norm selector reaches 13.00% on

Model quantization

Vocabulary-row quantization proxies

0 significant wins vs random

ARGUS

production process

6/6

papers completed

640

campaign-hours

576

Engineer rounds

286

Reviewer revisions

89

session rolls

16

Stage rollbacks